

VICHRUTH M

Machine Learning Engineer | Computer Vision | Edge AI & Model Compression | AI Automation

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PROFESSIONAL SUMMARY

Machine Learning and Computer Vision engineer (B.Tech Computer Science, VIT Vellore, 2028) specialising in model compression, quantization, and edge AI deployment. Research internships at NIT Tiruchirappalli and IIIT Kottayam; first-author paper pending publication in an IEEE journal. Achieved 92.0% Rank-1 accuracy on the Market-1501 person re-identification benchmark, shipped a fully offline multimodal AI system on Raspberry Pi, and published an open-source model-compression toolkit to PyPI. Designed and deployed a production B2B AI automation pipeline for a paying freelance client.

TECHNICAL SKILLS

- **Languages:** Python (advanced), C++ (intermediate), SQL, JavaScript, Java, Kotlin
- **Machine Learning / Deep Learning:** PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, XGBoost, LoRA/PEFT, Vision Transformers (ViT), TransReID, CLIP, LSTM, transfer learning, supervised fine-tuning, mixed-precision training, gradient accumulation
- **Computer Vision:** OpenCV, ONNX Runtime, YOLO11-pose, MediaPipe, YuNet/SFace face recognition, FAISS vector search, person re-identification, object detection, image classification
- **Edge AI & Model Compression:** FP16/INT8/4-bit quantization, model pruning, TinyML, llama.cpp, Raspberry Pi deployment, Android on-device inference
- **LLM & AI Engineering:** RAG, prompt engineering, structured output, embeddings, vector search, OpenAI API, small language models (Phi), offline/on-premise LLM deployment
- **Backend, Data & Automation:** FastAPI, Flask, Django, Node.js, Express, REST APIs, PostgreSQL, SQLite, Next.js, n8n, Stripe API, Telegram Bot API, Google Sheets API, Docker, Git, Linux

WORK EXPERIENCE

Summer Research Intern, LLM & Multimodal AI — National Institute of Technology, Tiruchirappalli | Jun 2026 – Jul 2026

- Designed and built a fully offline edge AI classroom attentiveness monitoring system on Raspberry Pi 8 GB under Dr. B. Janet, combining computer vision with a 4-bit quantized local LLM and zero cloud dependency.
- Built the face detection and recognition pipeline (OpenCV YuNet, SFace 128-D embeddings, cosine similarity) and a multimodal YOLO11-pose + MediaPipe engagement layer (gaze, drowsiness, hand-raise, emotion), reaching ~78% attentive-classification accuracy.
- Benchmarked the edge deployment against a high-compute cloud VLM baseline to quantify the accuracy–latency trade-off; evaluated face-detection backbones (MediaPipe, YOLOv8n-face, UltraFace) for 70-student overhead single-camera classrooms.
- Shipped the full stack solo: Flask admin portal with live monitoring, automated SMTP reporting, and remediation of XSS/CSRF vulnerabilities; findings feeding a journal paper in preparation.

Summer Research Intern, Computer Vision & Model Compression — Indian Institute of Information Technology, Kottayam | May 2026 – Jun 2026

- Extended TransReID (ViT-B/16) with a five-branch multi-task architecture — identity head, three soft-biometric attribute branches, and a domain adversarial head via Gradient Reversal Layer — reaching **92.0% Rank-1 accuracy on Market-1501** under Dr. Sridhar Raj S.
- Diagnosed and resolved three silent architectural failures (tensor-shape extraction, evaluation feature mismatch, missing identity supervision) that suppressed Rank-1 to 0.3% without producing crashes or abnormal loss curves, by validating intermediate tensor state directly.
- Ran systematic ablations across multi-task attribute learning, orthogonal part pooling, and triplet + BNNeck training entirely within a 6 GB VRAM budget using FP16 mixed precision and gradient accumulation; first-author paper pending publication in an IEEE journal.

Freelance AI & Automation Engineer — Independent Client Projects, Remote | 2025 – Present

- Designed, built, and deployed an 8-phase B2B lead-generation and enrichment pipeline for an Australian migration-services client on self-hosted n8n: automated scraping, GPT-4o-mini website audits, personalised outreach generation, reply sentiment classification, proposal generation, Stripe payments, and onboarding across 15+ API integrations.
- Architected the system as decoupled event-driven phases (schedule, webhook, and payment triggers) rather than a chained pipeline, so a single phase failure cannot cascade and a human approval gate sits between every AI-generated output and the customer.
- Built the operator control dashboard in Next.js (deployed on Vercel) with an approve/edit/skip review queue and an LLM regeneration loop; loaded and enriched 5,832 scraped leads.

App Developer — Padasalai, Tiruppatur (Remote) | Nov 2024 – Jan 2026

- Developed and maintained the web application for Padasalai, one of Tamil Nadu's largest educational platforms, building quiz and study tools serving 10th and 12th standard students across the state.
- Extended the platform with a Kotlin/Jetpack Compose Android app and a React + TypeScript admin dashboard on Firebase, shipped across 12 tagged releases with signed, R8-obfuscated builds; recognised with an award from the Padasalai and Sura Publications leadership.
- Modelled Firestore so a subject's full chapter tree costs a single document read, enforced access control through Firestore Security Rules rather than client-side routing, and built an AI import pipeline that parses uploaded PDF/DOCX question papers into structured, human-reviewed questions.

Machine Learning Intern — KalkiNi AI Surveillance Systems, Vellore | Feb 2025 – Sep 2025

- Built multi-threaded video-stream data pipelines (NumPy, Pandas) achieving a 35% preprocessing speedup; trained and fine-tuned deep learning classifiers for real-time threat profiling on live CCTV feeds using multi-task transfer learning with shared feature extraction.

AI Intern — Cloudstier Solutions Pvt. Ltd., Tiruppatur | May 2025 – Jun 2025

- Deployed an on-premise NLP chatbot (Django + Microsoft Phi small language model); owned supervised fine-tuning, prompt alignment, structured output formatting, and REST endpoint validation with Postman.

PROJECTS

ShrinkLens — Model Compression Toolkit (published on PyPI) github.com/vichruth/ShrinkLens

- Published an open-source Python package and CLI that quantizes vision models across FP32 → FP16 → INT8 → INT4 and reports measured model size, accuracy, and inference latency at each precision level, benchmarked on a real Vision Transformer.

InboxIQ — Student Email Copilot (Browser Extension + FastAPI Backend)

- Built and deployed a Manifest V3 browser extension (Chrome and Firefox) with a FastAPI backend that triages any email on demand: LLM-based categorisation, personalised importance scoring, deadline extraction into browser reminders, and a legitimacy check on opportunity emails.
- Designed a bring-your-own-key architecture so each user's requests run on their own free LLM quota and the server never stores credentials; combined unit-tested deterministic red-flag rules with LLM reasoning, and made the provider swappable between a cloud API and a fully local model via a single env variable.

NeuroLog — Offline Semantic Video Search github.com/vichruth/NeuroLog

- Zero-shot CLIP + FAISS video search engine running fully offline in 6 GB VRAM with ~0.284 s query latency over a 1,414-frame index; manual FP16 weight casting cut GPU memory from ~1.5 GB to ~600 MB with retrieval accuracy intact.

EdgeID — Offline Person Re-Identification on Android

- Trained a MobileNetV3-Large embedding network with batch-hard triplet loss (PyTorch, Market-1501) into a sub-6 MB on-device model; built a two-stage detector + 256-D embedding + nearest-neighbour pipeline in a Kotlin Android app running fully offline.

NIKI-LAUDA-AI — Real-Time Race Engineering Pipeline

- Decodes 50 Hz binary C-struct telemetry over UDP sockets from a physics simulator; training a PyTorch LSTM on sliding-window time-series data for predictive tyre-degradation modelling.

PUBLICATIONS

- **First-author paper, pending publication, IEEE journal:** "Person Re-Identification using Multi-Task Vision Transformers with Domain Adversarial Training" — Vichruth M, Dr. Sridhar Raj S, IIIT Kottayam, 2026.
- Second research paper in preparation with Dr. B. Janet (NIT Tiruchirappalli) on edge multimodal classroom engagement detection.

EDUCATION

B.Tech, Computer Science and Engineering — Vellore Institute of Technology (VIT), Vellore | Jul 2024 – Jul 2028

- Relevant coursework: Data Structures & Algorithms, Artificial Intelligence, Database Management Systems, Computer Architecture & Organisation, Theory of Computation, Probability & Statistics, Digital Systems Design, Microprocessors.

Senior Secondary (CBSE) — Maharishi International Residential School, Chennai | 85% | 2024

CERTIFICATIONS, LEADERSHIP AND HONOURS

- **Certifications:** Machine Learning Specialization (DeepLearning.AI / Coursera) | Summer Internship Programme Certificate, NIT Tiruchirappalli | VIT Generative AI Bootcamp | AI to Vision (IIIT Kottayam) | App Development Workshop (IIT Madras)
- **Leadership:** Junior Core Member, The Electronics Club, VIT (TinyML, embedded C++ mentoring) | Campus Ambassador, Masters' Union
- **Honours:** 2nd Prize, Hackademia, graVITas '25 (FFCS-Buddy) | 2nd Runner-Up, National Ideathon 2024 (Pre-Cog Brakes) | Competitor, FREUID Challenge 2026 (IJCAI-ECAI) | Kaggle March ML Mania 2026